

UNIT 3

BONES AND MUSCLES

Learning objectives:

- Learn the structure and function of the skeleton and its main parts.
- Identify joint types (hinge and ball-and-socket) and their functions in movement.
- Understand types of muscles (voluntary, involuntary, cardiac) and their roles.
- Recognize the roles of tendons and ligaments in connecting bones and muscles.
- Create joint models to demonstrate movement and explain how joints function.

Our body is made up of bones and muscles. Together, they form a system that helps us stand, move, and do many activities. Bones provide structure, while muscles help with movement.

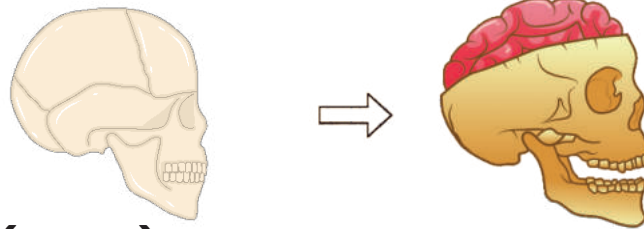
3.1 The Skeleton

What is a Skeleton? The skeleton is a framework of 206 bones in our body. It gives shape, support, and protection to our organs. Without bones, our body would be like jelly!



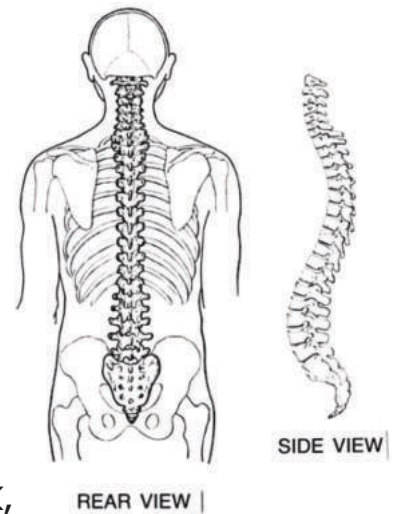
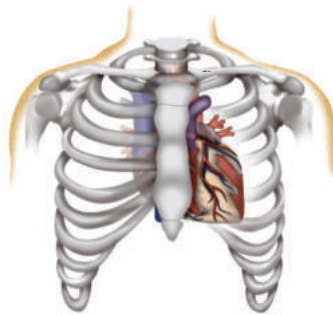
Main Parts of the Skeleton:

Skull: Protects our brain.

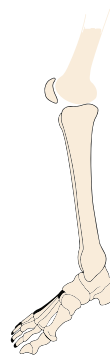
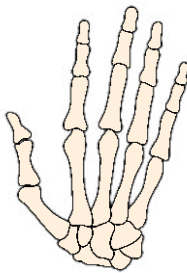


Backbone (Spine): Supports our back and helps us stand upright.

Ribcage: Protects our heart and lungs.



Limbs: Arms and legs help us to lift, walk, run, and do many other activities.



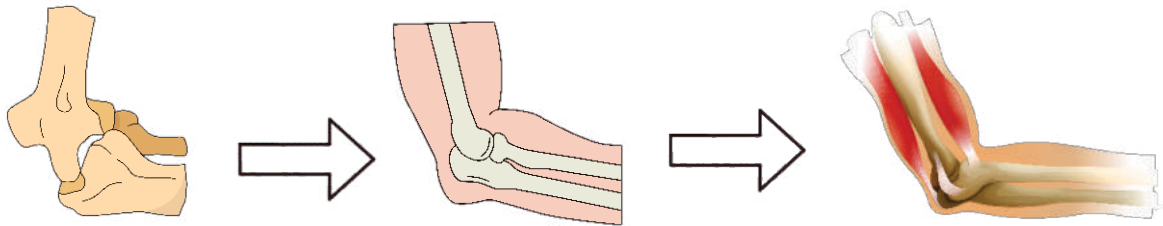
3.2 Joints and Movement

What are Joints? Joints are the places where two or more bones meet. They allow our body to bend and move. **For example,** our elbows and knees are joints that help our arms and legs bend.

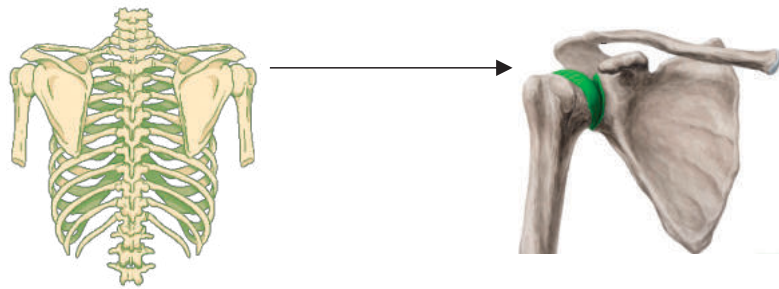


Types of Joints:

Hinge Joints: Allow back-and-forth movement like a door hinge (elbows, knees).



Ball-and-Socket Joints: Allow movement in many directions (shoulders, hips).



3.3 Muscles

What are Muscles? Muscles are soft stretchy tissues that are attached to bones. They help us move by pulling on the bones.

How Muscles Work: When you want to move your arm, your brain sends a signal to the muscles in your arm, and they pull the bones to make the arm move.

Types of Muscles:

Voluntary Muscles: Muscles you can control, like in your arms and legs. You use them when you want to pick up a toy or kick a ball.

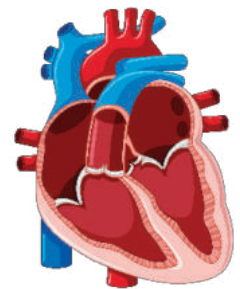
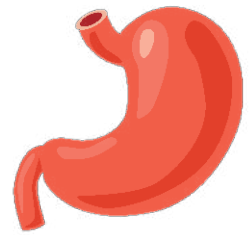
Example: Skeletal muscles.



Involuntary Muscles: Muscles that work without us thinking, like those in our stomach and heart. They help in digesting food and making our heart beat.

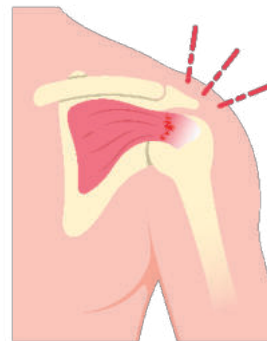
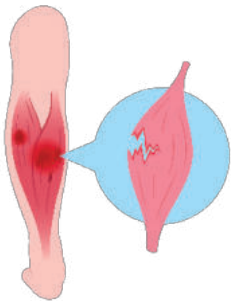
Example: Smooth muscles.

Cardiac Muscles: They are special muscles found only in the heart. They keep pumping blood throughout our body all the time.

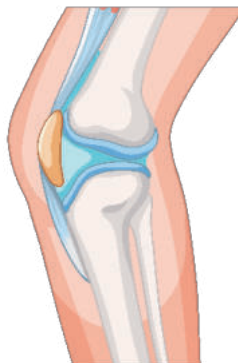


3.4 Tendons and Ligaments

Tendons: Strong, flexible cords that attach muscles to bones.



Ligaments: Tough tissues that connect bones to each other, holding joints together.



Fill in the blanks.

1. The skeleton is a framework of _____ bones in our body.
2. The _____ protects our brain.
3. _____ are places where two or more bones meet.
4. _____ muscles are muscles you can control, like those in your arms and legs.
5. _____ connect bones to each other, holding joints together.



Tick (✓) for True and cross (x) for False statements.

1. The ribcage protects our heart and lungs. ()

2. Ball-and-socket joints allow movement only in one direction. ()

3. Cardiac muscles are found throughout the body. ()

4. Tendons attach muscles to bones. ()

5. There are 300 bones in an adult human skeleton. ()



Multiple Choice Questions.

1. Which part of the skeleton protects the brain?

- a) Ribcage
- b) Skull
- c) Spine
- d) Limbs

2. What type of joint is found in the shoulder?

- a) Hinge Joint
- b) Ball and Socket Joint
- c) Fixed Joint
- d) Sliding Joint

3. Which of the following muscles work without us thinking about it?

- a) Voluntary Muscles
- b) Cardiac Muscles
- c) Skeletal Muscles
- d) All of the above

4. Ligaments connect:

- a) Bones to muscles
- b) Bones to bones
- c) Muscles to tendons
- d) Muscles to other muscles



Test Your Understanding

Q1: What is the main purpose of the skeleton?

Ans: _____

Q2: Explain the difference between hinge joints and ball-and-socket joints?

Ans: _____

Q3: What role do tendons play in the movement of our body?

Ans: _____

Q4: Describe the types of muscles and give an example of each?

Ans: _____



Q5: How does the ribcage help protect our body?

Ans: _____

Q6: What are joints?

Ans: _____

Q7: What are muscles?

Ans: _____



Critical Thinking

Imagine you are holding a heavy bag. What muscles are working, and what bones are supporting you?





Create a Simple Model of a Joint



ACTIVITY

- Modeling clay or playdough
- Two plastic straws or sticks (for bones)
- Rubber bands (for ligaments)
- Small plastic ball (for a ball-and-socket joint option)

Instructions:

1. Hinge Joint Model: Use two plastic straws to represent bones. Attach them with a rubber band at one end so they can only move back and forth, simulating a hinge joint (like the elbow or knee).

2. Ball-and-Socket Joint Model: Attach a plastic ball to the end of one straw using modeling clay or playdough. Insert this ball into a "socket" of clay on another straw to simulate the movement in many directions (like the shoulder or hip).

3. Explain: To your classmates or family members how each joint allows for different types of movement. Observe how the structure of each joint limits or enables certain movements.



ACTIVITY

Label the skeleton

foot bone

hip bone

leg bone

arm bone

neck bone

knee bone

thigh bone

shoulder bone

skull

ribs

backbone

hand bone

